1 Conservation of the Optimal-Frontier Defect

Let $\mathcal{A}$ be the output alphabet and let P, Q be probability laws on $\mathcal{A}$ satisfying

$$P = Q.$$

For $n \geq 0$, let $\mathcal{K}_n$ denote the set of probability laws on $Z_{\geq 0}$ which occur as roots of exact depth-n zero-repair trees, and define

$$\mathcal{B}_n := \inf_{B \in \mathcal{K}_n} B, \qquad \mathcal{B}_0 := 0.$$

An exact depth-n zero-repair tree is a family

$$\{B_h\}_{|h| \leq n}$$

satisfying, at every internal node h,

$$Q * B_h = \sum_{x \in \mathcal{A}} P(x)\, \delta_x * B_{hx}. \tag{1}$$

For a node h define its remaining horizon

$$r(h) := n - |h|.$$

Since the subtree rooted at h is itself an exact depth-$r(h)$ zero-repair tree,

$$B_h \in \mathcal{K}_{r(h)}.$$

We therefore define the *optimal-frontier defect* at h by

$$\Delta_h := B_h - \mathcal{B}_{r(h)}. \tag{2}$$

Necessarily

$$\Delta_h \geq 0.$$

For a word

$$g = (g_1, \ldots, g_s) \in \mathcal{A}^s$$

write

$$P(g) := \prod_{j=1}^{s} P(g_j).$$

[Multiscale conservation of the optimal-frontier defect] Let

$$\{B_h\}_{|h| \leq n}$$

be any exact depth-n zero-repair tree, and let h be any node with remaining horizon

$$r := r(h).$$

Then for every integer

$$0 \le s \le r$$

one has the exact identity

$$\sum_{g\in\mathcal{A}^s} P(g)\,\Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}. \tag{3}$$

Equivalently,

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{g\in\mathcal{A}^s} P(g)\Delta_{hg} - \Delta_h. \tag{4}$$

In particular, for one generation,

$$\sum_{x\in\mathcal{A}} P(x)\Delta_{hx} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-1}. \tag{5}$$

We first prove conservation of the expected reserve.

Taking expectations in the exact zero-repair identity eq:ZR gives

$$Q + B_h = \sum_{x\in\mathcal{A}} P(x)\big(x + B_{hx}\big).$$

Hence

$$Q + B_h = P + \sum_{x\in\mathcal{A}} P(x)B_{hx}.$$

Since

$$P = Q,$$

the common mean cancels, and therefore

$$B_h = \sum_{x\in\mathcal{A}} P(x)B_{hx}. \tag{6}$$

Iterating eq:mean-conservation for s generations yields

$$B_h = \sum_{g\in\mathcal{A}^s} P(g)B_{hg}. \tag{7}$$

Every descendant hg at distance s has remaining horizon $r-s$. By definition of the frontier defect,

$$B_{hg} = \mathcal{B}_{r-s} + \Delta_{hg}.$$

Substituting this into eq:sstep-mean gives

$$B_h = \sum_{g\in\mathcal{A}^s} P(g)\big(\mathcal{B}_{r-s} + \Delta_{hg}\big).$$

Since

$$\sum_{g\in\mathcal{A}^s} P(g) = 1,$$

we obtain

$$B_h = \mathcal{B}_{r-s} + \sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg}. \tag{8}$$

On the other hand, again by definition,

$$B_h = \mathcal{B}_r + \Delta_h.$$

Comparing with eq:descendant-expansion,

$$\mathcal{B}_r + \Delta_h = \mathcal{B}_{r-s} + \sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg}.$$

Rearranging proves

$$\sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.$$

Taking $s = 1$ gives

$$\sum_x P(x)\Delta_{hx} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-1}.$$

[Frontier contact] A node h is called a *frontier contact* if

$$\Delta_h = 0.$$

Equivalently,

$$B_h = \mathcal{B}_{r(h)}.$$

Since $B_h \in \mathcal{K}_{r(h)}$, a frontier contact is precisely a node whose subtree is optimal for its remaining horizon.

[Counterfactual conservation law] Let h be a frontier contact with remaining horizon r, and let

$$g_* \in \mathcal{A}^s$$

be a descendant history such that hg_* is again a frontier contact.

Then

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{g \in \mathcal{A}^s g \neq g_*} P(g)\Delta_{hg}. \tag{9}$$

Thus the contact branch g_* carries zero frontier defect, while the entire increase in optimal bill between horizons $r-s$ and r is carried by the alternative branches.

Because h is a frontier contact,

$$\Delta_h = 0.$$

Theorem 1 therefore gives

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg}.$$

Since hg_* is also a frontier contact,

$$\Delta_{hg_*} = 0.$$

Its contribution vanishes, leaving

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{g \neq g_*} P(g) \Delta_{hg}.$$

Define the marginal cost of one additional horizon level by

$$d_r := \mathcal{B}_r - \mathcal{B}_{r-1}.$$

[Exact export of the marginal bill] Suppose that h is a frontier contact of remaining horizon r and that one child hx_* is again a frontier contact.

Then

$$d_r = \sum_{x \neq x_*} P(x) \Delta_{hx}. \tag{10}$$

In particular, whenever $d_r > 0$, at least one sibling of the contact child has strictly positive frontier defect.

Equation eq:onestep-conservation gives

$$\sum_x P(x) \Delta_{hx} = \Delta_h + d_r.$$

Since

$$\Delta_h = \Delta_{hx_*} = 0,$$

we obtain

$$d_r = \sum_{x \neq x_*} P(x) \Delta_{hx}.$$

All terms are nonnegative, so if $d_r > 0$ at least one of them is strictly positive.

[Frontier-defect martingale] Let

$$H_t = (X_1, \ldots, X_t), \qquad X_1, X_2, \ldots \overset{\text{iid}}{\sim} P,$$

be a random branch of an optimal depth-n tree.

Then

$$\Delta_{H_t} = \mathcal{B}_n - \mathcal{B}_{n-t}. \tag{11}$$

Moreover,

$$Z_t := \Delta_{H_t} - \big(\mathcal{B}_n - \mathcal{B}_{n-t}\big) \tag{12}$$

is a martingale.

For a node H_t with remaining horizon $n-t$, Theorem 1 with $s = 1$ gives

$$[\Delta_{H_{t+1}} \mid H_t] = \Delta_{H_t} + \mathcal{B}_{n-t} - \mathcal{B}_{n-t-1}.$$

Hence

$$[Z_{t+1} \mid H_t] = \Delta_{H_t} + \mathcal{B}_{n-t} - \mathcal{B}_{n-t-1} - \big(\mathcal{B}_n - \mathcal{B}_{n-t-1}\big).$$

The deterministic terms simplify:

$$\mathcal{B}_{n-t} - \mathcal{B}_{n-t-1} - \mathcal{B}_n + \mathcal{B}_{n-t-1} = -\mathcal{B}_n + \mathcal{B}_{n-t}.$$

Therefore

$$[Z_{t+1} \mid H_t] = \Delta_{H_t} - \big(\mathcal{B}_n - \mathcal{B}_{n-t}\big) = Z_t.$$

Thus (Z_t) is a martingale.

At the optimal root,

$$\Delta = 0,$$

so

$$Z_0 = 0.$$

Consequently

$$Z_t = 0,$$

which gives

$$\Delta_{H_t} = \mathcal{B}_n - \mathcal{B}_{n-t}.$$

Interpretation

Theorem 1 establishes an exact multiscale conservation law.

Below a node with remaining horizon r, there are $|\mathcal{A}|^s$ possible histories of length s. Nevertheless their frontier defects are constrained by the single scalar identity

$$\sum_{g \in \mathcal{A}^s} P(g)\Delta_{hg} = \Delta_h + \mathcal{B}_r - \mathcal{B}_{r-s}.$$

Thus the exponentially many possible futures do not carry independent optimality costs.

They form a decomposition of one conserved frontier defect.

Most strikingly, if an optimal subtree contains a descendant which is itself optimal for the shorter remaining horizon, then

$$\Delta_{hg_*} = 0,$$

and therefore

$$\mathcal{B}_r - \mathcal{B}_{r-s} = \sum_{g \neq g_*} P(g)\Delta_{hg}.$$

The realised optimal continuation carries no excess cost at all. The full marginal cost of maintaining the longer horizon is stored in the alternative continuations.

In this precise mathematical sense, the bill is a property of the ensemble of possible futures rather than of the realised history alone.

Frontier-defect conservation principle.

At every scale, an exact zero-repair tree conserves the expected bill. After subtracting the optimal bill appropriate to the remaining horizon, this becomes an exact transport law for the nonnegative frontier defect.

Whenever one continuation remains on the optimal frontier, the entire marginal cost of extending the horizon is necessarily exported into the counterfactual sibling branches.